CNENGR202 Instrumentation and Control

Part 2 – L3 Temperature

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Mercury/Alcohol thermometer - 1



- Relatively large bulb of mercury or ethanol is attached to thin tube. As liquid heats up it expands causing the liquid to 'climb' up tube.

- Thermal expansion coefficients:
  - Mercury $182 \times 10^{-6}\ K^{-1}$
  - Ethanol $750 \times 10^{-6}\ K^{-1}$

- E.G Ethanol filled ball with radius of 3mm attached to 0.5mm inner thickness tube ($\pi R^2 = \mathbf{3.1416} \times \left(\mathbf{0.5}/\mathbf{2}\right)^2 = \mathbf{0.196}\ \text{mm}^2$ cross section).

- Ball volume = $\frac{\mathbf{4}}{\mathbf{3}}\pi R^3 = \frac{\mathbf{4}}{\mathbf{3}} \times \mathbf{3.1416} \times \mathbf{3}^3 = \mathbf{113.1}\ \text{mm}^3$

- A rise of 10℃ results in an expansion of $\mathbf{10} \times \mathbf{750} \cdot \mathbf{10}^{-6} \times \mathbf{113.1} = \mathbf{0.848}\ \text{mm}^3$

- This corresponds to an advancement up the tube of $\mathbf{0.848}\,/\,\mathbf{0.196} = \mathbf{4.33}\ \text{mm}$

- If filling Mercury ?

Mercury/Alcohol thermometer - 2



- Thermal expansion coefficients:
  - Mercury $182 \times 10^{-6}\ K^{-1}$

- Mercury filled ball with radius of 3mm attached to 0.5mm inner thickness tube ($\pi R^2 = \mathbf{3.1416} \times \left(\mathbf{0.5}/\mathbf{2}\right)^2 = \mathbf{0.196}\ \text{mm}^2$ cross section).

- Ball volume = $\frac{\mathbf{4}}{\mathbf{3}}\pi R^3 = \frac{\mathbf{4}}{\mathbf{3}} \times \mathbf{3.1416} \times \mathbf{3}^3 = \mathbf{113.1}\ \text{mm}^3$

- A rise of 10℃ results in an expansion of $\mathbf{10} \times \mathbf{182} \cdot \mathbf{10}^{-6} \times \mathbf{113.1} = \mathbf{0.2058}\ \text{mm}^3$

- This corresponds to an advancement up the tube of $\mathbf{0.2058}/\mathbf{0.196} = \mathbf{1.05}\ \text{mm}$

Thermal emittance (emissivity)

- Refers to a materials ability to release absorbed heat. Expressed as a number between 0 and 1 and is the ratio of the energy emitted from a surface to the energy emitted by a black surface.

- Emittance depends on Temperature, Wavelength, Electronic structure, Angle of viewing

Optical pyrometer - 1

- Measures the thermal radiation (irradiance, J in W/m^2), from an object
- Through the Stefan–Boltzmann law (σ - Stefan- Boltzmann constant, ε - emissivity), the target temperature T can be determined.

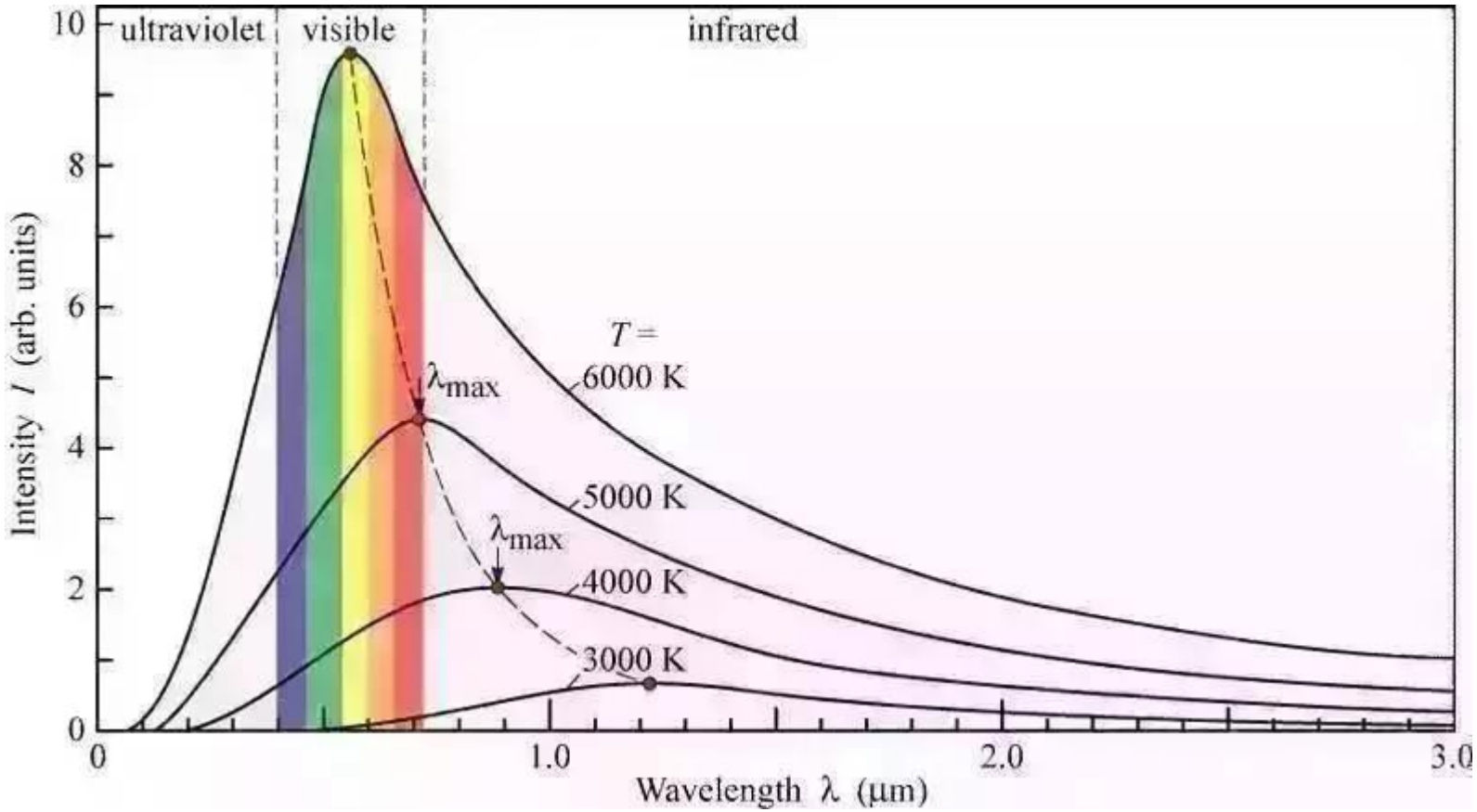
$$J = \varepsilon\sigma T^4 \qquad T = \sqrt[4]{\frac{J}{\varepsilon\sigma}} \qquad \sigma = 5.67\times10^{-8} Wm^{-2}K^{-4}$$

- Depending on surface type, the emissivity coefficient can alter greatly (Black paint 0.93, Concrete 0.85, Sand 0.76, Stainless steel weathered 0.85, Stainless steel polished 0.08, polished brass 0.03).



Optical pyrometer - 2

- Colour pyrometer
- Measure ratio of intensity in two close narrow frequency bands



Resistance Temperature Detectors (RTD's) - 1

- Exploit change in electrical resistance of some materials with temperature.
- Usually made of platinum, often called Platinum Resistance Thermometers (PRTs) useful for applications below 600℃.
- *Type 1: Film thermometers* have a thin layer (~1μm) of platinum on a substrate (Low cost and fast response but different expansion rates of the substrate and platinum can give 'strain gauge' effects and stability problems).
- *Type 2: Wire-wound thermometers* have greater accuracy, especially for wide temperature ranges. The coil diameter provides a compromise between mechanical stability and allowing expansion of the wire to minimize strain and consequential drift.



Resistance Temperature Detectors (RTD's) - 2

- Usually nominal resistance of 100 ohms at 0℃ (Pt100 sensors).

$$R_T = R_0\left[1 + AT + BT^2 + CT^3\left(T - 100\right)\right] \qquad \text{-200℃–0℃}$$

$$R_T = R_0\left[1 + AT + BT^2\right] \qquad \text{0℃–650℃}$$

- Measuring range -- Pt10

-200℃–850℃

- R_T is the resistance at temperature T, R_0 is the resistance at 0℃, constants typically

$$A = 3.9083\times10^{-3}\ {}^0\text{C}^{-1} \qquad B = -5.775\times10^{-7}\ {}^0\text{C}^{-2} \qquad C = -4.183\times10^{-12}\ {}^0\text{C}^{-3}$$

- B and C are small thus changes almost linearly with temperature.

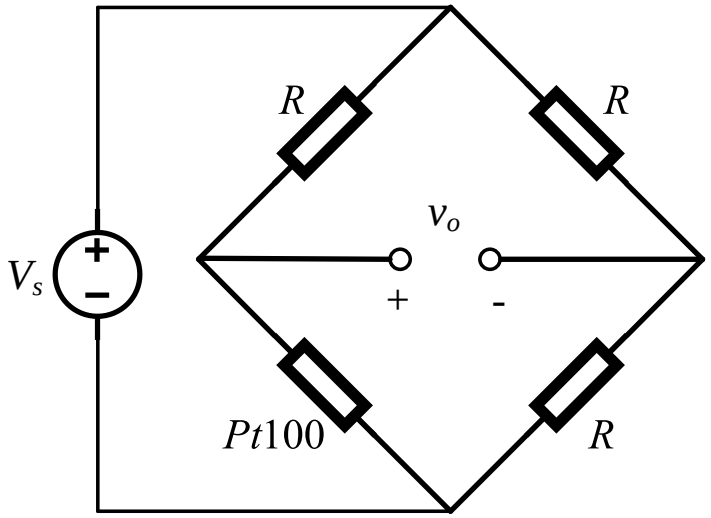
Indexing table

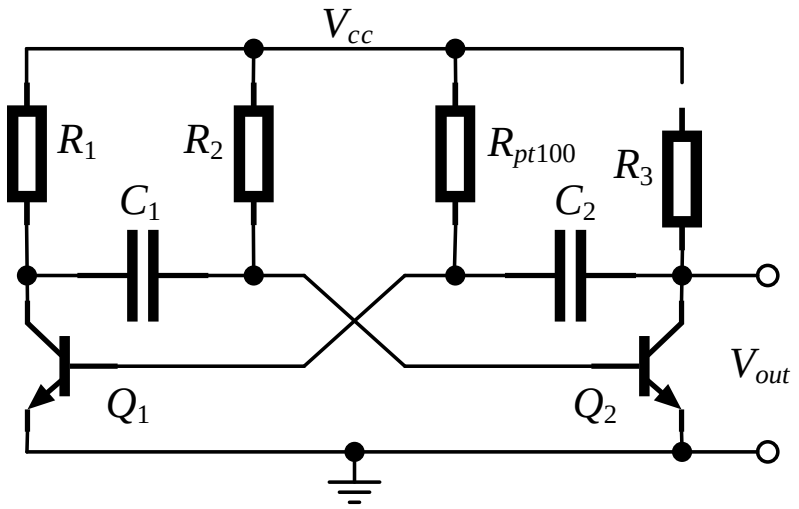
Thermometric Characteristics of Pt100 Sensor

T[℃]	R[Ω]
-200	18.52
-100	60.26
-50	80.31
0	100
50	119.4
100	138.51
150	157.33
200	175.86

T[℃]	R[Ω]
250	1
300	212.05
400	247.09
500	280.98
600	313.71
700	345.28
800	375.70
850	390.48

Mearing circuit





Thermistor-1

- Resistance varies with temperature but differ from RTD's in that they are generally a ceramic or polymer, while RTDs use pure metals.

- The temperature response is also different; RTDs are useful over larger temperature ranges, while thermistors typically achieve a higher precision within a limited temperature range (−50 °C to 130 °C).

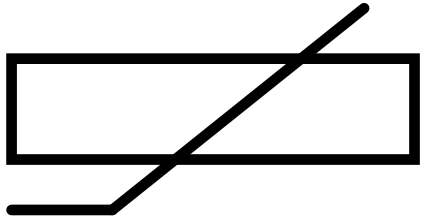
- PTC: Positive Temperature Coefficient
  NTC: Negative Temperature Coefficient
  CTR: Critical Temperature Resistor

Thermistor-2

- Exact numbers vary from component to component, but it is important to remember the general relationship



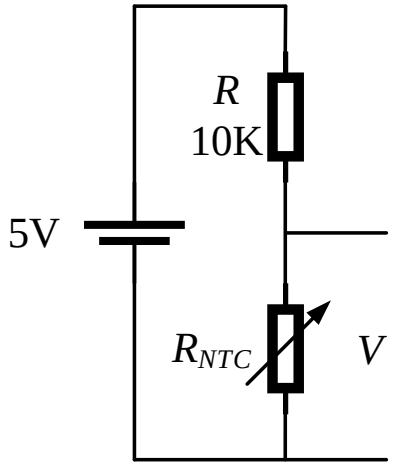
- NTC $R = R_0 \exp^{B/T} \longrightarrow T = \dfrac{B}{\ln\left(R/R_0\right)}$
- PTC $R = R_0 \exp^{BT} \longrightarrow T = \dfrac{\ln\left(R/R_0\right)}{B}$



Thermistor symbol

Class question

- A thermistor is used to calculate rough temperature in a small furnace.
- Design a potential divider circuit which looks like this



- If the thermistor has a B value of 3435 and a R_{25} value of 10KΩ.
- The voltmeter V has a reading 1.0V
- What is the temperature?

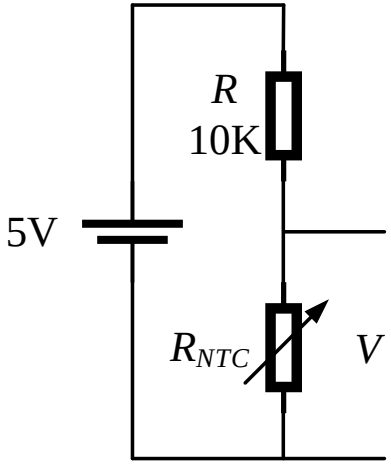
Hint

$$R = R_0 \exp^{B/T}$$

Answer

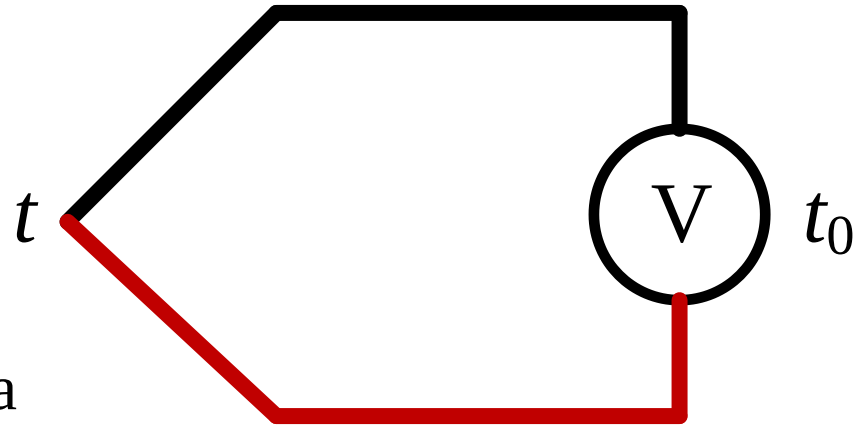
$$1=\frac{R_t}{R_t+10000}\times 5 \longrightarrow R_t+10000=5R_t$$

$$R_t=\frac{10000}{4}=2500$$



$$\begin{cases} R_{25}=R_0\exp^{B/(273.15+25)} \\ R_t=R_0\exp^{B/(273.15+t)} \end{cases} \longrightarrow \begin{cases} 10000=R_0\exp^{3435/295.15} \\ 2500=R_0\exp^{3435/(273.15+t)} \end{cases} \longrightarrow t=59.9°$$

Seebeck effect (塞贝克效应)/ thermocouples



- EMF(ElectroMotive Force) is created in the presence of a temperature difference between two different metals.
- Relationship between temperature difference and output voltage of a thermocouple is nonlinear, approximated by a polynomial usually implemented in a digital controller

$$T = \sum_{n=0}^{N} a_n V^n = a_0 V^0 + a_1 V^1 + a_2 V^2 + a_3 V^3 + \ldots etc$$

K-type (inexpensive, common, 73K-1625K, ~41 μV/K)

E-type (Relatively low temperature, ~68 μV/K)

J-type (230K-1030K , ~55 μV/K)

N-type (Very high temperatures, ~39 μV/K)

Parameters of Industrial Thermocouples

Type	*Metal* (+)	*Metal* (–)	k_T [μV/°C]	*Temperature Range* [°C]	*Tolerance* ΔT at 750°C	*Properties*
K	Chromel Ni-Cr	Nickel	40.5	–270 to +1,370	Class 1: ±3.0°C; class 2: ±5.6°C	Good linearity
S	Platinum	Pt-Rh10 90%Pt+10%Rh	6.4	–50 to +1,760	Class 1: ±1°C; class 2: ±1.9°C	Good accuracy
J	Iron	Copper–nickel	51.7	–210 to +1,200	Class 1: ±3°C; class 2: ±5.6°C	Not expensive, good sensitivity
T	Copper	Copper–nickel	49	–200 to +400	Class 1: ±1.4°C; class 2: ±2.6°C	Not expensive, good sensitivity
W3	W-Re3 97%W+3%R	W-Re25: 75%W+25%Re	18	0 to +2,320	No class 1; class 2: ±10°C	High operation temperature

Indexing table of thermocouples

K型热电偶分度表

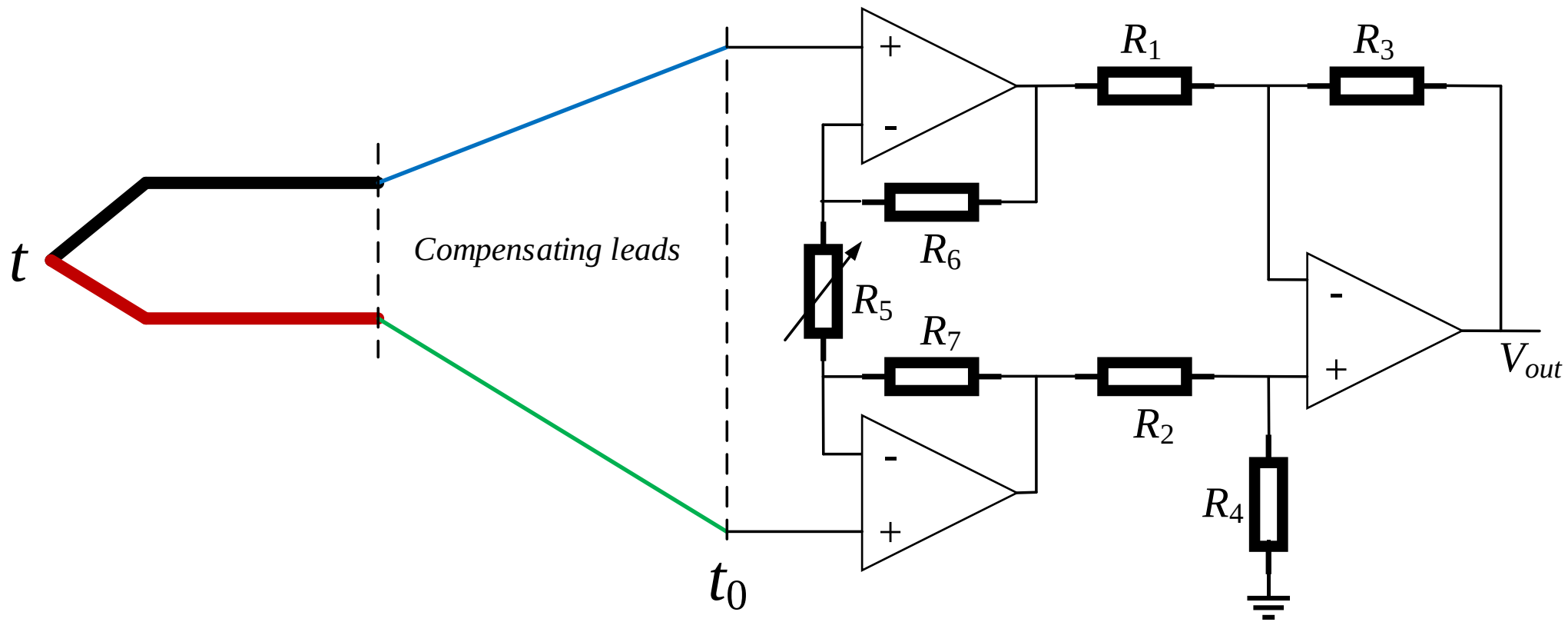
温度（℃）	0	1	2	3	4	5	6	7	8	9
	热电动势mV									
0	0	0.0397	0.0794	0.1191	0.1588	0.1985	0.2382	0.2779	0.3176	0.3573
10	0.397	0.4371	0.4772	0.5173	0.5574	0.5975	0.6376	0.6777	0.7178	0.7579
20	0.798	0.8385	0.879	0.9195	0.96	1.0005	1.041	1.0815	1.122	1.1625
30	1.203	1.2438	1.2846	1.3254	1.3662	1.407	1.4478	1.4886	1.5294	1.5702
40	1.611	1.6521	1.6932	1.7343	1.7754	1.8165	1.8576	1.8987	1.9398	1.9809
50	2.022	2.0634	2.1048	2.1462	2.1876	2.229	2.2704	2.3118	2.3532	2.3946
60	2.436	2.4774	2.5188	2.5602	2.6016	2.643	2.6844	2.7258	2.7672	2.8086
70	2.85	2.8916	2.9332	2.9748	3.0164	3.058	3.0996	3.1412	3.1828	3.2244
80	3.266	3.3075	3.349	3.3905	3.432	3.4735	3.515	3.5565	3.598	3.6395
90	3.681	3.7224	3.7638	3.8052	3.8466	3.888	3.9294	3.9708	4.0122	4.0536
100	4.095	4.1363	4.1776	4.2189	4.2602	4.3015	4.3428	4.3841	4.4254	4.4667
1300	52.398	52.4329	52.4678	52.5027	52.5376	52.5725	52.6074	52.6423	52.6772	52.7121
1310	52.747	52.7816	52.8162	52.8508	52.8854	52.92	52.9546	52.9892	53.0238	53.0584
1320	53.093	53.1276	53.1622	53.1968	53.2314	53.266	53.3006	53.3352	53.3698	53.4044
1330	53.439	53.4733	53.5076	53.5419	53.5762	53.6105	53.6448	53.6791	53.7134	53.7477
1340	53.782	53.8163	53.8506	53.8849	53.9192	53.9535	53.9878	54.0221	54.0564	54.0907
1350	54.125	54.1591	54.1932	54.2273	54.2614	54.2955	54.3296	54.3637	54.3978	54.4319
1360	54.466	54.5001	54.5342	54.5683	54.6024	54.6365	54.6706	54.7047	54.7388	54.7729
1370	54.807	54.8411	54.8752	54.9093	54.9434	54.9775	55.0116	55.0457	55.0798	55.1139

K 型 镍铬-镍硅 热电偶 E（t）分度表 第 1 页

温度 ℃	0	1	2	3	4	5	6	7	8	9
	热电动势 mV									
-50	-1.889	-1.925	-1.961	-1.996	-2.032	-2.067	-2.102	-2.137	-2.173	-2.208
-40	-1.527	-1.563	-1.600	-1.636	-1.673	-1.709	-1.745	-1.781	-1.817	-1.853
-30	-1.156	-1.193	-1.231	-1.268	-1.305	-1.342	-1.379	-1.416	-1.453	-1.490
-20	-0.777	-0.816	-0.854	-0.892	-0.930	-0.968	-1.005	-1.043	-1.081	-1.118
-10	-0.392	-0.431	-0.469	-0.508	-0.547	-0.585	-0.624	-0.662	-0.701	-0.739
-0	-0	-0.039	-0.079	-0.118	-0.157	-0.197	-0.236	-0.275	-0.314	-0.353
0	0.000	0.039	0.079	0.119	0.158	0.198	0.238	0.277	0.317	0.357
10	0.397	0.437	0.477	0.517	0.557	0.597	0.637	0.677	0.718	0.758
20	0.798	0.838	0.879	0.919	0.960	1.000	1.041	1.081	1.122	1.163
30	1.203	1.244	1.285	1.326	1.366	1.407	1.448	1.489	1.530	1.571

500	20.644	20.687	20.730	20.772	20.815	20.857	20.900	20.943	20.985	21.028
510	21.071	21.113	21.156	21.199	21.241	21.284	21.326	21.369	21.412	21.454
520	21.497	21.540	21.582	21.625	21.668	21.710	21.753	21.796	21.838	21.881
530	21.924	21.966	22.009	22.052	22.094	22.137	22.179	22.222	22.265	22.307
540	22.350	22.393	22.435	22.478	22.521	22.563	22.606	22.649	22.691	22.734
550	22.776	22.819	22.862	22.904	22.947	22.990	23.032	23.075	23.117	23.160

Mearing circuit



Method 6: Gas thermometer

- Measures temperature by the variation in volume or pressure of a gas.
- One common apparatus is a constant volume thermometer consisting of a bulb connected by a capillary tube to a manometer.



Ideal gas law: $pV=nRT$

Charles's Law: $p_1V_1 = p_2V_2$

p is the absolute pressure of the gas (pa)

V is the volume of the gas (m^3)

n is the number of moles of gas present

R is the gas constant (8.314 $JK^{-1}mol^{-1}$)

T is the absolute temperature (k).

Guide to what approach to use for what temperature range

Temperature range	Primary fixed points (approx. values)	Interpolating instrument	Method of Interpolation
0.65K to 5K		Gas thermometer	Measurement of vapour pressures of Helium 3 and 4
3K to 24.6K	b.p. of H_2(13.8K) b.p. of Neon(24.6K)	Gas thermometer	Constant volume gas thermometer
14K to 303K	b.p. of H_2(13.8K) b.p. of Neon(24.6K) b.p. of Oxygen(54.4K) b.p. of Argon(83.8K) b.p. of Gallium(303K) b.p. of Water(273.16K)	Platinum resistance thermometer	Specified reference function together with a deviation equation whose coefficients are determined in the calibration against fixed points.
0.1℃ to 962℃	m.p. of Gallium(30℃) f.p. of Indium(157℃), Tin(232℃), Zinc(420℃), Al.(660℃), Silver(962℃)	Platinum resistance thermometer	As above
>962℃	f.p. of Silver(962℃) f.p. of Gold(1064℃) f.p. of Copper(1084℃)	Radiation pyrometer	Defined by Planck's law taking the radiation emitted from a black body cavity at fixed points